SUPPORTING ADULT GUIDANCE

Formulae from words and diagrams

Formulae and Rearranging · Lesson 3. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson turns descriptions in words and diagrams into formulae, and then uses them. Learners tap each phrase of a description to see it become part of a bar model and part of a formula, write formulae for perimeters and areas by walking round shapes, meet word-order traps such as $2x + 3$ against $2(x + 3)$, and use three gym formulae to decide which is cheapest and when the answer changes.

Driving Question: *How do I turn a description into a formula?*

The mathematics, in plain terms

A linear formula of the form total = fixed amount + rate × quantity models many real situations; the rate multiplies the variable and the fixed amount does not. Word order maps to operation order, with brackets used where an addition must happen before a multiplication or power. A perimeter is the sum of all side lengths; like terms are then collected.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — From words to a formula, then using it

A plumber charges £40 to call out, plus £25 an hour. Write a formula for the cost C of h hours, and find the cost of 2½ hours.

1. Fixed part: paid once.	40
2. 'For each' part: £25 for each hour.	25h
3. Put them together.	C = 40 + 25h
4. Substitute h = 2½: multiply first.	C = 40 + 25 × 2.5 = 40 + 62.5 = 102.5

C = 40 + 25h, and 2½ hours cost £102.50.

Example 2 — From a diagram to a formula

A rectangle is $(x + 2)$ cm long and x cm wide. Write a formula for its perimeter P.

1. Go round once, writing every side. $P = (x + 2) + x + (x + 2) + x$
2. Collect the x terms. $x + x + x + x = 4x$
3. Collect the numbers. $2 + 2 = 4$

$P = 4x + 4$.

Check the formula against the words

Learners' formulae are checked by value, so any correct form is accepted (3 + 2m, 2m + 3, 3 + m × 2). When a formula is wrong, the lesson shows a short table comparing it with the description. Encourage learners to test their own formula with one simple value before checking — it is the habit that catches most errors.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Words to bars to formula: tap each phrase, watch the bar model grow, then write and use the formula.	~14 min
Movement	Walk round the shape: perimeter and area formulae from labelled diagrams.	~9 min
Reflection	Word-order traps: the same words in a different order give a different formula.	~8 min
Creativity	Which gym is cheaper? Write three formulae, then use them to decide.	~9 min
Rest	Spotting fixed and 'for each' parts in everyday prices.	~2 min
Nutrition	Intellectual: modelling a real decision with mathematics.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Swapping the fixed amount and the rate (2 + 3m instead of 3 + 2m).
Ask: 'Which amount do you pay even for 0 miles?'

Multiplying the fixed amount as well ((3 + 2)m).
Draw the bar for 0, 1 and 2 miles: does the £3 repeat?

'5 less than n' written as 5 – n.
Try n = 10: is 5 less than 10 equal to 5 – 10?

Missing or double-counting sides of a perimeter.
Walk round the shape with a finger, saying each side aloud once.

Rounding a fractional input (2½ hours taken as 2).
Ask the learner to write 2½ as 2.5 before substituting.

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: writing simple expressions, and substituting values from Lesson 2. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner writes C = 2 + 3m for 'a taxi costs £3 plus £2 for every mile'. The aim is a question that lets them find the slip themselves.

- "How much would a journey of 0 miles cost?"
- "What does your formula give for 1 mile? What should it be?"

- “Which number is paid again for every mile?”
- “Can you draw the bar for 2 miles?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Does the order of terms matter, for example 3 + 2m or 2m + 3?

No. Both are correct, and both are accepted. The convention of putting the letter term first is common, but not required here.

Why include a decreasing example?

Savings that go down each week show that the 'for each' part can be subtracted: M = 60 – 4w.

Why is Gym C never cheapest?

Its joining fee is higher than Gym A's (which has none) and its monthly rate is higher than Gym B's. It is a useful discussion point about modelling.

What comes next?

Lesson 4 derives formulae from patterns and shapes, and explains why they work.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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